Comprehensive Guide to Large Language Models (LLMs)

The Transformer Architecture Large Language Models are based on the Transformer architecture introduced by Vaswani et al. in 2017. The architecture relies on the Self-Attention mechanism, which computes representations of input sequences by focusing on different parts of the same sequence.

Popular LLM Families State-of-the-art LLMs include: - OpenAI GPT: Generative Pre-trained Transformer series (e.g., GPT-4o) trained on large datasets using autoregressive objectives. - Google Gemini: Multimodal native models (Gemini 1.5/2.5 Pro and Flash) capable of parsing text, audio, images, and video. - Anthropic Claude: High-reasoning model family (Claude 3.5 Sonnet) designed around constitutional AI for safety. - Open-Weights Models: Meta's Llama series, Mistral, and Google's Gemma models which can be deployed locally.

Fine-Tuning and Parameter-Efficient Adaptation Fine-tuning adapts a pre-trained model to specific tasks. Parameter-Efficient Fine-Tuning (PEFT) methods, like Low-Rank Adaptation (LoRA), insert small, trainable matrices into Transformer layers while freezing the base weights, reducing VRAM needs.